IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-56

DC Charging System(BEVC-DC001)-II

Dr. Apurv Kumar Yadav

Department of Electrical Engineering

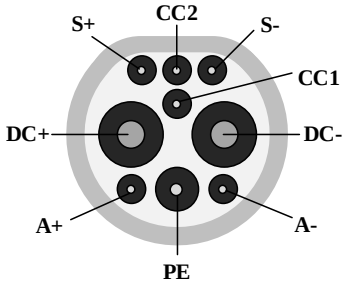


Recap

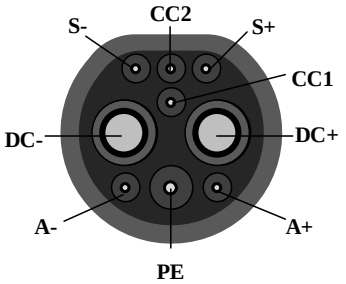
- DC001 DC charging system
- CAN Communication

Recap: BEVC-DC001

- Input: 3-phase AC
- Output: Supports up to a maximum of 10kW for 48V system and 15 kW for 72V systems
- Communication is derived from IEC 61851-24B and GB/T 27930 (application layer)
- Uses CAN communication
- GB/T 20234.3 Connector with 5 m Cable



Charger Plug



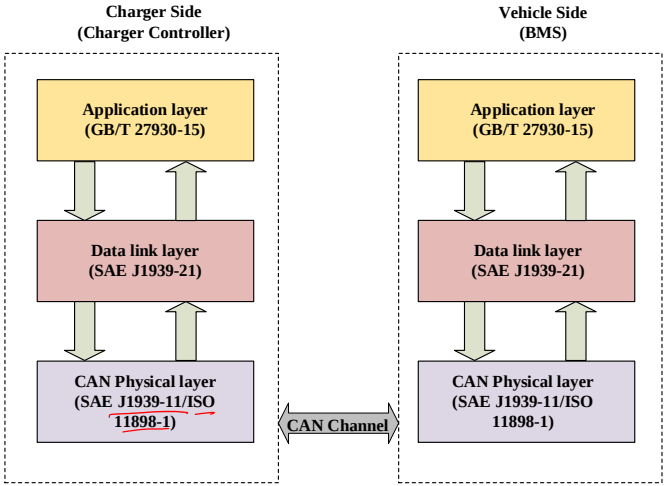
Vehicle inlet connector

Introduced by Department of Heavy Industries, Government of India

Source: https://heavyindustries.gov.in/sites/default/files/2023-09/Standardization%20of%20protocol.pdf

Recap: Communication in DC001 (Derived from GB/T 27930)

- GB/T 27930-2015 defines rules for implementing the physical and data link layers based on SAE J1939 standard.
- The application layer of GB/T 27930-2015 defines representation of parameters and parameter group numbers.
- Speed – 250Kbps, uses CAN 2.0B
- Only two participants at a time – a charger and a BMS in the vehicle – with fixed addresses:
  - 86 (56h) for the charger
  - 244 (F4h) for the BMS
- No direct connections to other CAN systems in the vehicle, such as the power train CAN.
- CAN data frames are transmitted in the form of Protocol Data Unit (PDU) defined in GB/T 27930.



Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

Recap: CAN Communication

- The controlled Area Network (CAN) is a two-wire serial bus communication and messages are sent sequentially.
- Message-based protocol also allows for several features to be added via software alone.
- CAN based communication is widely used in automotive industries, avionics systems, medical systems and escalators.
- Advantages: low cost, centralized communication, robust, efficient, error capability, flexible, multi-master configuration
- Lower the Message ID higher will be priority.

CAN → CAN message

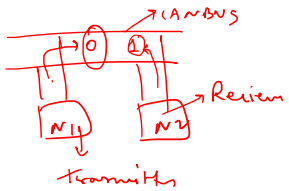
Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

CAN Communication Electrical Properties

- CAN communication is a two wire system
  - CANH
  - CANL
- The data is sent differentially on CANH and CANL channels. The difference between the CANH and CANL voltages gives the actual data.
  - Eliminates corruption of data due to electromagnetic interference from nearby circuits
- CAN communication has two forms of logic
  - Logic'0' is called as Dominant logic
  - Logic'1' is called Recessive logic
- The differential voltage (V_{diff}) is defined as

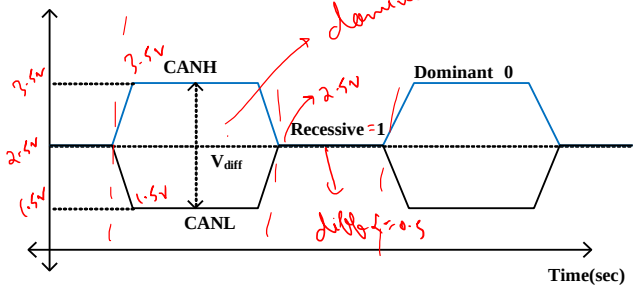
$$V_{diff} = CANH - CANL$$

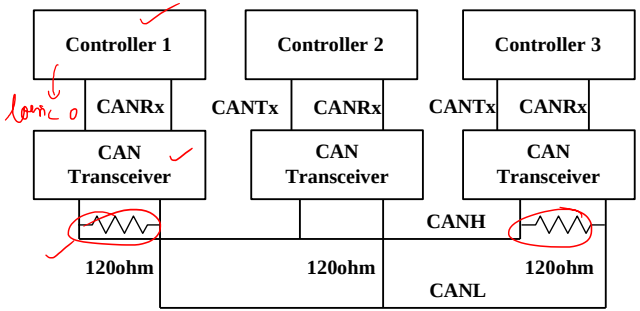
- The dominant logic is detected when $V_{diff} >= 0.9V$
- The recessive logic is detected when $V_{diff} <= 0.5V$

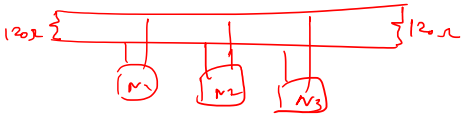


CAN Communication Electrical Properties Cont.

- When logic ‘0’ (dominant) is applied, a pulse of 3.5V will come on CANH, while a pulse of 1.5V on CANL. Similarly, in logic ‘1’ (recessive), the CAN bus will remain at 2.5V.
- The CANH and CANL are shielded twisted pair, which avoids any interference to nearby circuit as well as makes the communication more reliable from interference due to nearby circuits.
- Two terminating resistors of 120Ω each are used at each end of the bus. Terminating resistors are used to match the impedance of the network. This helps to mitigate signal integrity issues like large fall or rise time.

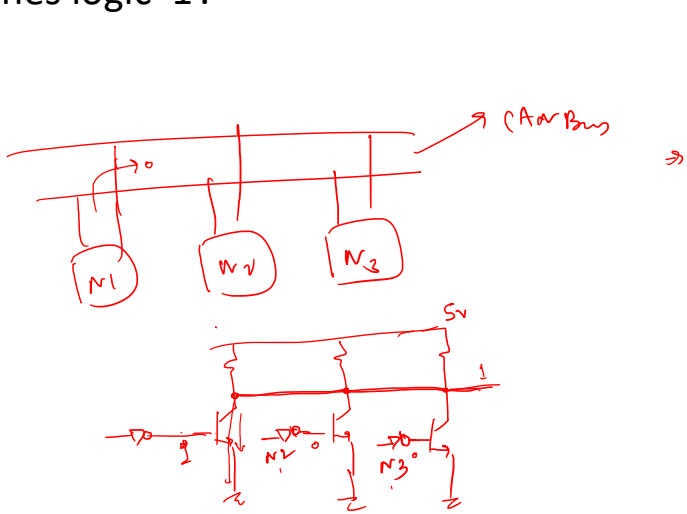






CAN Communication Electrical Properties Cont.

- The nodes works like open collector configuration: they can pull down the CANBUS to dominant logic ‘0’. But cannot make it to make it ‘1’.
- If any one node pushes ‘0’ logic, it makes the entire bus logic to be ‘0’. However, the bus has logic ‘1’ when all the nodes pushes logic ‘1’.



N1	N2	N3	Can Bus logic
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

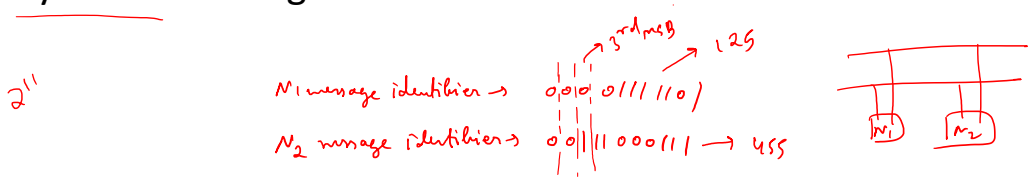
Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

CAN Communication: Data Frame CAN 2.0A

- The data frame for CAN 2.0A communication comprises of

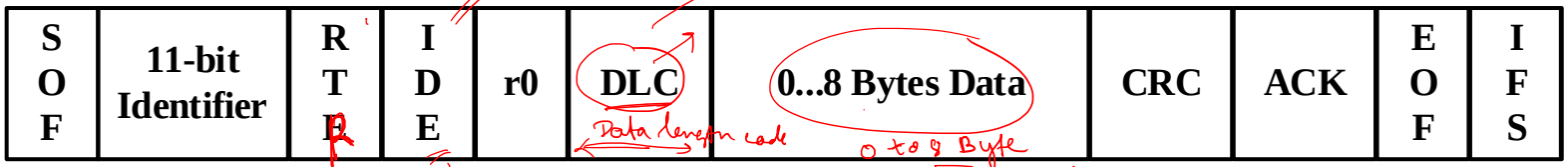
SOF	11-bit Identifier	RTR	IDE	r0	DLC	0...8 Bytes Data	CRC	ACK	EOF	IFS

- **SOF**: start of frame bit to synchronize nodes
- **Identifier**: Establishes the priority of the messages. The lower the binary value, the higher will be the priority of the message.



Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

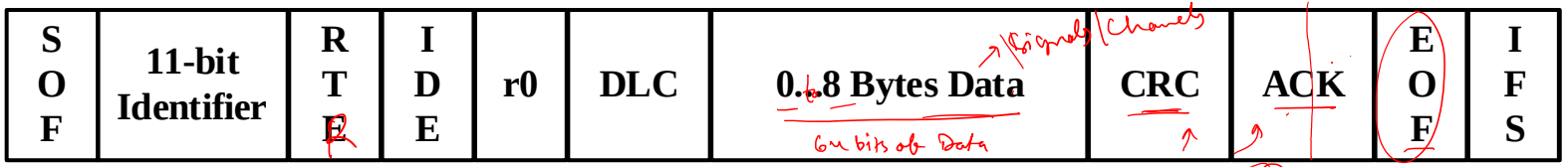
CAN Communication: Data Frame CAN 2.0A Cont.



- **RTR**: The single remote transmission request (RTR) bit indicates whether the frame is a data frame or a remote request frame. This bit is used to request data from another node on the CAN bus. When the RTR bit is dominant (0), the frame indicate data on the bus. When the RTR bit is recessive (1), the frame acts as a request for data from another node that has the matching message ID.
- **IDE**: A dominant single identifier extension (IDE) bit means that a standard CAN identifier with no extension is being transmitted.
- **r0**–Reserved bit
- **DLC**: 4-bit data length code that contains the number of bytes transmitted. Since the maximum of 8 bytes of data can be transmitted, thus, require 4-bit.

Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

CAN Communication: Data Frame CAN 2.0A Cont.



- **Data**: Up to 8 bytes of application data can be transmitted. The data can be 64 signals of 1 bit, 8 signals of 8 bit each, etc.
- **CRC**: Cyclic redundancy check for error detection. Mostly a 16-bit number.
- **ACK**: 2 bit with one as acknowledgement and other one is delimiter. Every node receiving an accurate message overwrites this recessive bit in the original message with a dominant bit, indicating an error-free message has been sent. Sender put recessive logic and receiver overwrites with dominant logic.
- **EOF**: End of Frame. It is a 7-bit field marks the end of a CAN frame (message).
- **IFS**: A 7-bit interframe space (IFS) contains the time required by the receiver to move the received frame to its buffer.

Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

CAN Communication: Data Frame CAN 2.0B

- The data frame for CAN 2.0B communication comprises of

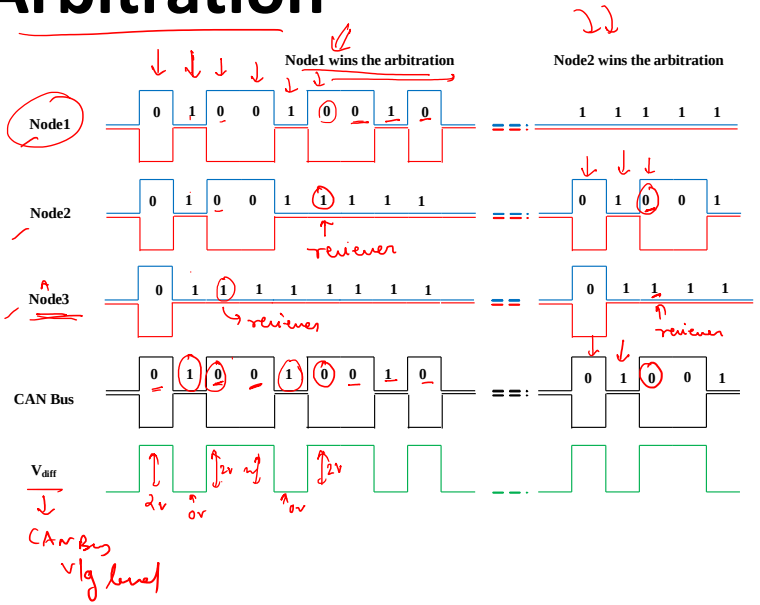


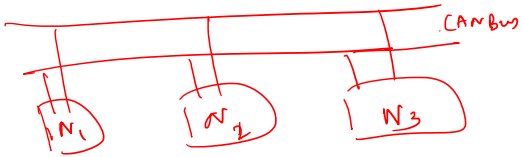
- **SRR**–The substitute remote request (SRR) bit. A recessive IDE bit indicates that the CAN message is in extended format. It allows nodes to recognize extended CAN frames without needing to know the full 29-bit identifier.
- **IDE**–A recessive bit in the identifier extension (IDE) indicates that more identifier bits follow. The 18-bit extension follows IDE.
- **r1**–reserved bit
- **Identifier**: Establishes the priority of the messages. The lower the binary value, the higher will be the priority of the message. In the case of CAN2.0B, It's an 29-bit identifier.

Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

CAN Communication: Arbitration

- If node 1,2 and 3 is trying to communicate with each other as shown below.
- Initially, node 1 wins the arbitration. Node 2 and node 3 will wait for some time and then start the communication and wins arbitration based on the lower most value.
- Since the common CAN bus is shared between the nodes, once any node puts the dominant bit or logic '0', it will win the arbitration and sends the signal while others wait.
- Once the communication is completed, nodes try to send the data again.





Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

CAN Database File

- These are the text files containing information of CAN messages, signals and their parameters
- Each CAN message can have multiple CAN signals.
- The device manufacturers usually provide the CAN database file along with their product. For each signal, CAN databases define the rules for conversion to engineering units.
- The following data is stored in databases:
  - Message name, Message ID and DLC bit
  - Channel/signal name
  - Location (start bit) and size of the channel in a message
  - Signal type
  - Byte order (Intel/Motorola): Little-endian or big-endian
  - Scaling factor and units string
  - Range (minimum or maximum)
  - offset

Source: https://www.ti.com/lit/an/sloa101b/sloa101b.pdf?ts=1759042551767

Thank You